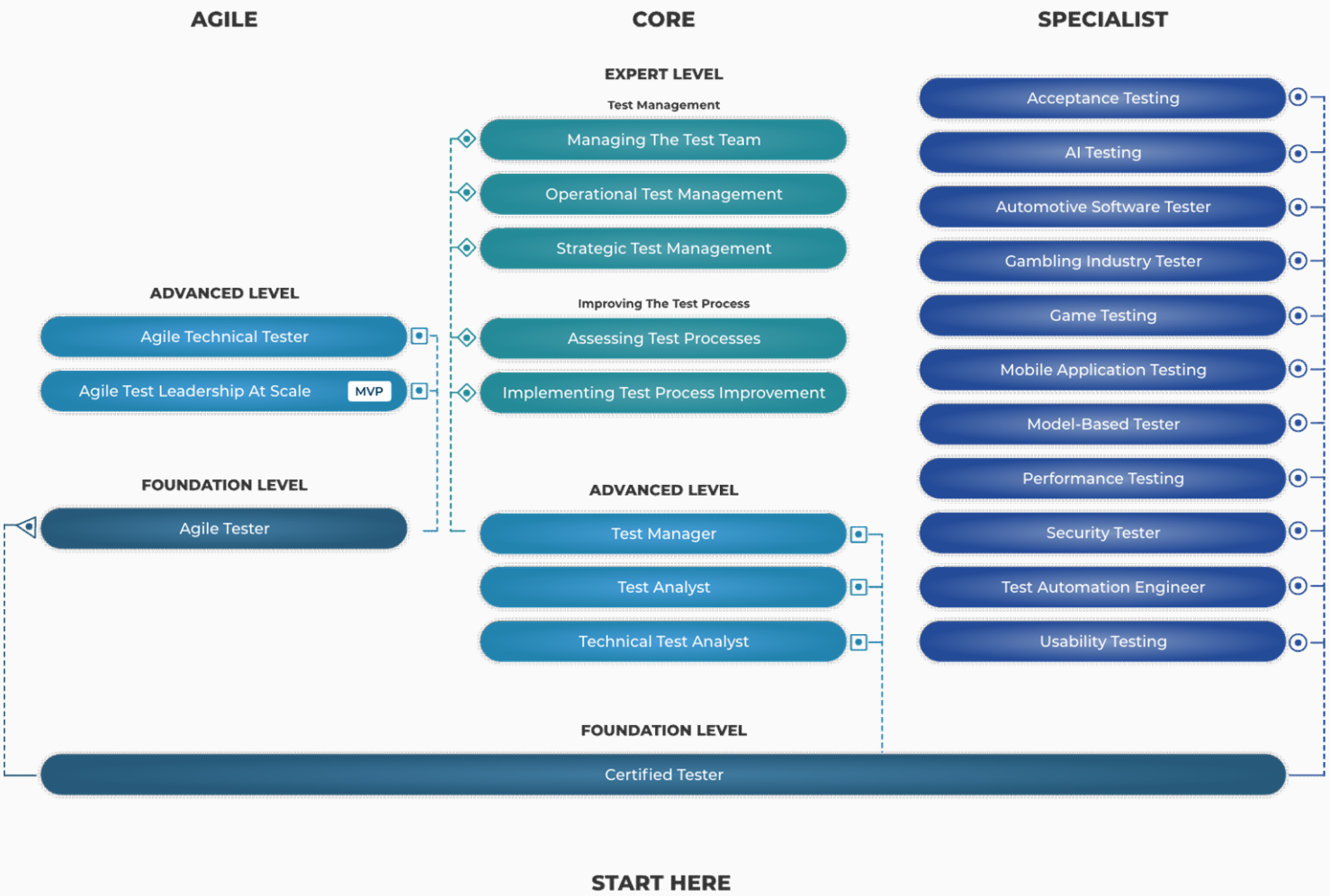
INTRODUCTION TO THE COURSE

ISTQB

- ISTQB (International Software Testing Qualifications Board)
- It offers many certificates in software testing

ISTQB website

ROAD MAP IN SOFTWARE TESTING DOMAIN



COURSE CONTENT

ISTQB® Certified Tester Foundation Level (CTFL)

Fundamentals of Testing	Testing Throughout the Software Development Lifecycle	Static Testing	Test Techniques	Test Management	Tool Support for Testing
What is Testing?	Software Development Lifecycles Models	Static Testing Basics	Categories of Test Techniques	Test Organization	Test Tool Considerations
Why is Testing Necessary?	Test Levels	Review Process	Black-box Test Techniques	Test Planning and Estimation	Effective Use of Tools
Seven Testing Principles	Test Types		White-box Test Techniques	Test Monitoring and Control	
Test Process	Maintenance Testing		Experience-based Test Techniques	Configuration Management	
The Psychology of Testing				Risks and Testing	
				Defect Management	

Preparing for the Exam

K-Level	Number of Questions	Question Timing in minutes	Total Time by K-Level (Approximate)
K1	8	1	8
K2	24	1	24
K3	8	3	24
K4	0	4	0
TOTALS	40		56

Exam Structure

Preparing for the Exam

- 40 single-choice questions from K1 to K4, require 26/40 (65%) to pass.
  - 08 questions of K1 (remember)
  - 24 questions of K2 (understand)
  - 08 questions of K3 (apply) and K4 (analyse)
- 75 minutes for non-native speakers
- No scores in the Certificate
- No Expiration

Preparing for the Exam

Questions per chapter

- C1: 08 questions
- C2: 05 questions
- C3: 05 questions
- C4: 11 questions
- C5: 09 questions
- C6: 02 questions

Audience

- The Foundation Level syllabus forms the basis of the International Software Testing Qualifications Board (ISTQB®) Certified Tester Scheme. The exam and the Syllabus
- ISTQB® Foundation Level is relevant across software delivery practices including Waterfall, Agile, DevOps and Continuous Delivery.
- Foundation Level qualification is suitable for anyone who needs to demonstrate practical knowledge of the fundamental concepts of software testing including people in roles such as testers, test analysts, test engineers, test consultants, test managers, user acceptance testers and software developers.

Business Outcomes

- Promote efficient and effective communication by using a common vocabulary for software testing.
- Understand fundamental concepts of software testing.
- Demonstrate understanding of how different development and testing practices, and different constraints on testing, may apply in optimizing testing to different contexts.
- Contribute effectively to reviews.
- Use established techniques for designing tests at all test levels.
- Interpret and execute tests from given test specifications. Report on test results.

Business Outcomes (cont.)

- Understand test management principles for resources, strategies, planning, project control and risk management
- Write and communicate clear and understandable defect reports
- Understand the project factors that drive the test priorities and test approach
- Understand the value that software testing brings to stakeholders
- Appreciate how testing activities and work products align with project objectives, measures and targets
- Assist in the selection and implementation process of testing tool

Learning Objectives

K1	remember, recognize, recall (15)
K2	understand, explain, give reasons, compare, classify, categorize, give examples, summarize (40)
K3	apply, use (7)
K4	analyze

Learning Objectives

The 2018 Foundation Level Learning objectives are as follows:

Chapter 1 Fundamentals of Testing

1.1 What is Testing?

LO-1.1.1 Identify typical objectives of testing (K1)
LO-1.1.2 Differentiate testing from debugging (K2)

1.2 Why is Testing Necessary?

LO-1.2.1 Give examples of why testing is necessary (K2)
LO-1.2.2 Describe the relationship between testing and quality assurance and give examples of how testing contributes to higher quality (K2)
LO-1.2.3 Distinguish between error, defect, and failure (K2)
LO-1.2.4 Distinguish between the root cause of a defect and its effects (K2)

1.3 Seven Testing Principles (K2)

LO-1.3.1 Explain the seven principles of testing (K2)

1.4 Test Process

LO-1.4.1 Explain the impact of context on the test process (K2)
LO-1.4.2 Describe the test activities and respective tasks within the test process (K2)
LO-1.4.3 Differentiate the work products that support the test process (K2)
LO-1.4.4 Explain the value of maintaining traceability between the test basis and the test work products (K2)

1.5 The Psychology of Testing (K2)

LO-1.5.1 Identify the psychological factors that influence the success of testing (K1)
LO-1.5.2 Explain the difference between the mindset required for test activities and the mindset required for development activities (K2)

Learning Objectives

Chapter 2 Testing Throughout the Software Development Lifecycle

2.1. Software Development Lifecycle Models

LO-2.1.1 Explain the relationships between software development activities and test activities in the software development lifecycle (K2)

LO-2.1.2 Identify reasons why software development lifecycle models must be adapted to the context of project and product characteristics (K1)

LO-2.1.3 Recall characteristics of good testing that are applicable to any life cycle model (K1)

2.2 Test Levels (K2)

LO-2.2.1 Compare the different test levels from the perspective of objectives, test basis, test objects, typical defects and failures, and approaches and responsibilities (K2)

2.3 Test Types (K2)

LO-2.3.1 Compare functional, non-functional and white-box testing (K2)

LO-2.3.2 Recognize that functional and structural tests occur at any test level (K1)

LO-2.3.3 Recognize that functional, non-functional and white-box tests occur at any test level (K1)

LO-2.3.4 Compare the purposes of confirmation testing and regression testing (K2)

2.4 Maintenance Testing (K2)

LO-2.4.1 Summarize triggers for maintenance testing (K2)

LO-2.4.2 Describe the role of impact analysis in maintenance testing (K2)

LO-2.4.3 Describe the role of impact analysis in maintenance testing (K2)

Learning Objectives

Chapter 3 Static Testing

3.1 Static Testing Basics

LO-3.1.1 Recognize types of software work product that can be examined by the different static testing techniques (K1)
LO-3.1.2 Use examples to describe the value of static testing (K2)
LO-3.1.3 Explain the difference between static and dynamic techniques, considering objectives, types of defects to be identified, and the role of these techniques within the software lifecycle (K2)

3.2 Review Process

LO-3.2.1 Summarize the activities of the work product review process (K2)
LO-3.2.2 Recognize the different roles and responsibilities in a formal review (K1)
LO-3.2.3 Explain the differences between different review types: informal review, walkthrough, technical review and inspection (K2)
LO-3.2.4 Apply a review technique to a work product to find defects (K3)
LO-3.2.5 Explain the factors that contribute to a successful review (K2)

Learning Objectives

Chapter 4 Test Techniques

4.1 Categories of Test Techniques

LO-4.1.1 Explain the characteristics, commonalities, and differences between black-box test techniques, white-box test techniques and experience-based test techniques (K2)

4.2 Black-box Test Techniques

LO-4.2.1 Apply equivalence partitioning to derive test cases from given requirements (K3)
LO-4.2.2 Apply boundary value analysis to derive test cases from given requirements (K3)
LO-4.2.3 Apply decision table testing to derive test cases from given requirements (K3)
LO-4.2.4 Apply state transition testing to derive test cases from given requirements (K3)
LO-4.2.5 Explain how to derive test cases from a use case (K2)

4.3 White-box Test Techniques

LO-4.3.1 Explain statement coverage (K2)
LO-4.3.2 Explain decision coverage (K2)
LO-4.3.3 Explain the value of statement and decision coverage (K2)

4.4 Experience-based Test Techniques

LO-4.4.1 Explain error guessing (K2)
LO-4.4.2 Explain exploratory testing (K2)
LO-4.4.3 Explain checklist-based testing (K2)

Learning Objectives

Chapter 5 Test Management

5.1 Test Organization

LO-5.1.1 Explain the benefits and drawbacks of independent testing (K2)
LO-5.1.2 Identify the tasks of a test manager and tester (K1)

5.2 Test Planning and Estimation

LO-5.2.1 Summarize the purpose and content of a test plan (K2)
LO-5.2.2 Differentiate between various test approaches (K2)
LO-5.2.3 Give examples of potential entry and exit criteria (K2)
LO-5.2.4 Apply knowledge of prioritization, and technical and logical dependencies, to schedule test execution for a given set of test cases (K3)
LO-5.2.5 Identify factors that influence the effort related to testing (K1)
LO-5.2.6 Explain the difference between two estimation techniques: the metrics-based technique and the expert-based technique (K2)

5.3 Test Monitoring and Control

LO-5.3.1 Recall metrics used for testing (K1)
LO-5.3.2 Summarize the purposes, contents, and audiences for test reports (K2)

Learning Objectives

5.4 Configuration Management

LO-5.4.1 Summarize how configuration management supports testing (K2)

5.5 Risks and Testing

LO-5.5.1 Define risk level by using likelihood and impact (K1)

LO-5.5.2 Distinguish between project and product risks (K2)

LO-5.5.3 Describe, by using examples, how product risk analysis may influence thoroughness and scope of testing (K2)

5.6 Defect Management

LO-5.6.1 Write a defect report, covering defects found during testing (K3)

Learning Objectives

Chapter 6. Tool Support for Testing

6.1 Test tool considerations

LO-6.1.1	Classify test tools according to their purpose and the test activities they support (K2)
LO-6.1.2	Identify benefits and risks of test automation (K1)
LO-6.1.3	Remember special considerations for test execution and test management tools (K1)

6.2 Test Planning and Estimation (K3)

LO-6.2.1	Identify the main principles for selecting a tool (K1)
LO-6.2.2	Recall the objectives for using pilot projects to introduce tools (K1)
LO-6.2.3	Identify the success factors for evaluation, implementation, deployment and on-going support of test tools in an organization (K1)

Taking the Exam

Multiple-Choice Test Taking Tips and Strategies:

#1: Read the entire question.

#2: Answer it in your mind first.

#3: Eliminate wrong answers.

#4: Use the process of elimination.

#5: Select the best answer.

#6: Read every answer option.

#7: Answer the questions you know first.

Taking the Exam

#8: Make an educated guess.

#9: It's usually best to stick with your first choice--but not always.

#10: "All of the above" and "None of the above"

#11: When there are seemingly two correct answers.

#12: Place your bet on the positive option.

#13: The more information... the better.

SUMMARY

- Preparing for the Exam
- Taking the Exam